

Deepak Natarajan

Staff QA Engineer

QA Automation | Playwright | Selenium | Python | API Automation | Team Leadership

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Professional Summary:

- Staff QA Engineer with 12+ years of experience in QA Automation, Enterprise Software Testing, and Quality Engineering across Marine, Aerospace, Semiconductor, ERP, and Broadcast domains. Expertise in automation frameworks, QA leadership, Python, Selenium, Robot Framework, API Automation, SQL, Jenkins, Git, and Agile methodologies. Developed AI-assisted validation solutions using LLMs to improve operational visibility, validation accuracy, and reduce manual verification effort.

Professional Experience:

Staff QA Engineer | ZeroNorth [Sept 2021 - Present],

- **Promoted to the role of Staff QA Engineer**, leading and mentoring a team of 5+ QA engineers to deliver high-quality automation and functional testing solutions.
- **Designed and implemented a reusable Excel-based validation tracker with the assistance of ChatGPT (LLM)** to automate day-wise Charter Party module validation, improving operational visibility, enhancing validation accuracy, accelerating issue identification, and significantly reducing manual verification effort.
- **Collaborated closely with Product Development Management (PDM) team and Development teams** to analyze business requirements, refine acceptance criteria, and align testing strategies with product objectives.
- **Designed and executed end-to-end test scenarios** for critical business workflows, ensuring stable software releases while minimizing production defects.
- **Revamped and stabilized critical application modules** by improving test coverage, strengthening validation processes, and proactively resolving recurring quality issues.
- **Supported multiple cross-functional teams** in delivering critical project milestones and production validation activities, ensuring smooth project execution and on-time releases.
- **Provided post-release production support**, performing production validation, root cause analysis, and timely issue resolution to ensure seamless deployments and business continuity.

Senior Test Engineer | Tata Elxsi [Dec 2018 — Aug 2021]

- **Developed and maintained scalable UI automation frameworks from scratch using Python and Robot Framework for both Web and Windows applications**, improving automation efficiency and test maintainability.
- **Analyzed business requirements** and designed comprehensive test scenarios, test cases, and test data to ensure complete functional and regression test coverage.
- **Developed and executed automation test scripts** for new features and maintained regression test suites, leveraging Jenkins for scheduled and continuous test execution.
- **Collaborated closely with developers**, business analysts, and cross-functional teams to identify, report, track, and resolve defects, ensuring timely delivery of high-quality software.
- **Configured and integrated** Python environments, supporting libraries, AutoIt, TortoiseGit, and Jenkins to establish and maintain a stable automation ecosystem.
- **Enhanced automation coverage** by designing reusable test components and improving framework maintainability, resulting in faster regression cycles and increased testing efficiency.

Test Engineer | HCL Technologies [July 2014 to Dec 2018],

- **Performed end-to-end functional and manual testing across Aerospace and Semiconductor domains**, ensuring software quality and compliance with business requirements.

- **Designed, reviewed, and executed** comprehensive test scenarios and test cases based on functional and technical specifications, ensuring complete test coverage.
- **Validated critical business workflows** and identified software defects through systematic functional, integration, and regression testing, contributing to stable and reliable software releases.
- **Collaborated with Business Analysts, Developers, and QA teams** during requirement analysis, test planning, execution, and release validation to ensure successful project delivery.
- **Performed GUI and backend database validation** to verify application functionality, data integrity, and overall system reliability.
- **Prepared test execution reports, defect metrics, Weekly Status Reports (WSRs), and quality dashboards** to provide stakeholders with clear project visibility and testing progress.
- **Managed the complete defect lifecycle**, including defect reporting, tracking, verification, and closure, while working closely with development teams to ensure timely resolution.

Technical Skills:

- Languages – **Python, SQL**
- Automation: **Playwright, Selenium, Robot Framework, PyTest**, Winium, AutoIt
- Version Control & CI/CD – **Git, GitHub, TortoiseGit, Gerrit and Jenkins**
- API Testing – **Postman, Python Requests and SoapUI**
- Tools - **VS Code, PyCharm and Liclipse, JIRA, qTest, TestLodge, Linear, PuTTY, Confluence and MTM**
- Methodologies - **Agile/Scrum**
- Operating Systems – **Windows & Linux (Basics)**
- Domain Expertise – **Marine, Broadcast, Semiconductor, Aerospace, ERP.**
- Testing Expertise – **Automation, Manual, API, Web, Desktop, Database and Mobile.**

Key Achievements:

- Received the “**BRAVO Award**” twice for delivering high-quality testing and exceptional project contributions.
- Honored with the “**ELXSIAN Award**” for exemplary performance, dedication, and project excellence.
- Received appreciation from clients and senior management for designing an **AI-assisted Excel-based validation tracker** for the Charter Party module using ChatGPT (LLM).

Core Strengths

- Strong analytical and problem-solving skills with proven ability to identify critical software defects and improve product quality.
- Effective team leader with experience mentoring QA engineers and driving quality initiatives.
- Strong stakeholder communication and collaboration skills across Product, Development, and QA teams.

Education

- **Master of Computer Applications (MCA)**
Osmania University – Suprabhath Institute for Management and Computer Studies | 2009 – 2012
- **Bachelor of Electronics Science (BES)**
Anna University – Kumararani Meena Muthiah College of Arts & Science | 2006 – 2009
- **Higher Secondary Education (XII) & Secondary Education (X)**
Sri Vidya Mandir Higher Secondary School | 2003 – 2005